

CSE4007 Artificial Intelligence - Assignment #2

Due: May 3rd, 2026 11:59 pm

The submission should include a code (both link to the colab *with full access permission* and .py format) and a report that has answers to the questions and results. Use **PyTorch**. Also, minimize the use of numpy. There will be mark deductions if numpy is used when PyTorch is preferable or should be used. Marks will be deducted if the submission does not include the requested files. **DO NOT** use other libraries, such as scikit-learn/sklearn, to use a model you are supposed to implement. **Using sklearn or any other third library or already built-in functions** that you are asked to implement **will result in 0 mark**. Implement **all the TODO sections** indicated in the assignment code. *Use this link to download the assignment code.*

Problem #1: Building DETRansformer (DETR)

- a) Implement the positional encoding of DETR (Sinusoidal encoding is required).
- b) Implement the attention module of DETR.
- c) Implement the encoder layers of DETR.
- d) Implement the decoder layers of DETR.
- e) Implement the heads (containing the box and class head) and the complete architecture of DETR.
- f) Describe the implementation verification process for each module and include relevant observations in your report. More precisely, you should explain why the code you wrote is correct in a way that convinces the reader of the report.

Problem #2: Building Loss Function for DETR

- a) Implement the Hungarian matcher.
- b) Implement the object loss.
- c) Implement the class loss.
- d) Implement the bounding box loss.

e) Describe the implementation verification process for each module and include relevant observations in your report. More precisely, you should explain why the code you wrote is correct in a way that convinces the reader of the report.

Problem #3: Achieving Best Performance

- a) Join *this competition*. To participate in this competition and receive a score, you must submit your predictions. For more detailed information and rules about the competition, please refer to the competition website.
- b) Develop the current framework of DETR to achieve the best performance as well as possible using any approaches and techniques. You can submit your predictions to receive a score in the competition mentioned above. Please make sure to refer to the competition rules for any prohibited actions regarding performance improvement.
- c) In your report, explain how you have achieved the best performance so far.